

TuniTech Abroad

Database Schema v1

Final validation draft before Service Contracts and implementation

Generated for the France-first IT job intelligence MVP

Architecture baseline: Django + PostgreSQL + Redis/Celery + OpenRouter + France Travail API

Schema style: normalized relational schema with controlled JSON usage

Public identifier strategy: internal BigAutoField IDs + UUID public_id for public routes

1. Document Purpose

This document defines the database schema for TuniTech Abroad MVP. It translates the BRD, PRD, and Technical Architecture into a Django/PostgreSQL-ready schema. It defines apps, models, fields, relationships, delete behavior, constraints, indexes, JSON fields, status enums, seed data, and migration order.

2. Schema Principles

- PostgreSQL is the source of truth. Redis is only for Celery broker, cache, rate limits, locks, and temporary counters.
- External job data is stored raw before normalization. RawJobRecord protects against API changes and allows replay of normalization.
- Final fit score is deterministic. LLM output supports extraction and explanation but is not the scoring authority.
- User-confirmed profile data is stronger than parsed CV data.
- PostgreSQL full-text search is used for job search from MVP. Avoid large icontains searches on descriptions.
- Privacy is first-class: CV deletion, account deletion, consent records, email opt-out, private CV storage.

3. Django Apps and Main Models

App	Models
accounts	User
profiles	CandidateProfile, ProfileSkill
cvs	CVUpload, CVParsedData
jobs	JobSource, IngestionRun, RawJobRecord, NormalizedJob, NormalizedJobSkill
skills	Skill, SkillAlias, UnmatchedSkillCandidate
matching	MatchResult, QuickMatchSession
recommendations	JobRecommendation, RecommendationRun, SavedJob
notifications	EmailPreference, EmailBatch, EmailEvent, EmailUnsubscribeToken
llm	PromptVersion, LLMCacheEntry, LLMUsageLog
privacy	ConsentRecord, DeletionRequest
analytics	UserEvent
core	SystemSetting optional

4. Shared Field and ID Conventions

Most business models include id, created_at, and updated_at. Use Django default BigAutoField for internal primary keys. For public-facing routes, expose UUID public_id instead of integer pk.

```
public_id = models.UUIDField(  
    default=uuid.uuid4,  
    unique=True,  
    editable=False,  
    db_index=True,  
)
```

Public routes must use public_id:
/jobs/<uuid:public_id>/
/matches/<uuid:public_id>/

Never expose internal URLs like:
/jobs/123/
/matches/456/

Public ID required on

- NormalizedJob
- CVUpload
- MatchResult
- QuickMatchSession

5. Common Enum Values

Enum	Values
current_level	student, internship_seeker, junior, mid_level, senior, unknown
target_type	summer_internship, standard_internship, pfe_end_of_study, first_job, junior_job, mid_level_job, unknown
language_level	unknown, A1, A2, B1, B2, C1, C2, native
remote_preference	remote, hybrid, on_site, any, unknown
relocation_preference	willing_to_relocate, remote_only, already_in_france, not_ready, unknown
job_status	active, stale, expired, removed, archived
job_type	internship, full_time_job, apprenticeship, contract, unknown
remote_type	remote, hybrid, on_site, unknown
parse_status	pending, processing, parsed, parsed_with_warnings, failed, deleted
llm_status	not_requested, pending, success, failed, cached, disabled

6. Accounts Schema

6.1 accounts.User

Custom User model from the beginning. Email is unique. Supports email/password, Google, GitHub, and admin accounts.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
username	CharField	no	Optional or auto-generated
email	EmailField	yes	Unique
first_name	CharField	no	Django default
last_name	CharField	no	Django default
password	CharField	no	Empty for pure social users
is_active	BooleanField	yes	Default true
is_staff	BooleanField	yes	Admin access
is_superuser	BooleanField	yes	Superuser access
date_joined	DateTimeField	yes	Auto
last_login	DateTimeField	no	Django default

Constraints:

- unique(email)

Indexes:

- email
- is_active
- is_staff
- last_login
- date_joined

Delete behavior:

- Controlled through privacy/account deletion flow.

7. Profiles Schema

7.1 profiles.CandidateProfile

Stores the candidate's editable/confirmed profile. This is the main source for matching.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	OneToOneField(User)	yes	One profile per user
full_name	CharField	no	Prefilled from auth/CV
phone	CharField	no	Candidate phone
location	CharField	no	Current location
linkedin_url	URLField	no	Manual field
github_url	URLField	no	Manual/OAuth-derived
portfolio_url	URLField	no	Portfolio
website_url	URLField	no	Personal site
current_level	CharField choices	yes	Default unknown
years_experience	DecimalField	no	Example: 1.5
target_country	CharField	yes	MVP default France
target_roles	JSONField	no	List of target roles for MVP
target_type	CharField choices	yes	Job/internship target
french_level	CharField choices	yes	Default unknown
english_level	CharField choices	yes	Default unknown
relocation_preference	CharField choices	yes	Default unknown
remote_preference	CharField choices	yes	Default unknown
profile_completeness_score	PositiveSmallIntegerField	yes	0-100
profile_completeness_status	CharField	yes	incomplete/usable/strong
is_confirmed	BooleanField	yes	User reviewed profile
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(user)

Indexes:

- user
- current_level
- target_country
- target_type
- profile_completeness_status
- updated_at

Implementation note:

- target_roles remains JSONField for MVP.
- If role matching or analytics become slow, migrate later to TargetRole and CandidateProfileTargetRole tables.

7.2 profiles.ProfileSkill

Stores canonical skills confirmed or inferred for a candidate profile. Matching should use this relational table, not raw JSON only.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
profile	ForeignKey(CandidateProfile)	yes	Candidate profile
skill	ForeignKey(Skill)	yes	Canonical skill
source	CharField	yes	user/cv_regex/cv_llm/oauth_github/admin

Field	Type	Required	Notes
confidence	DecimalField	no	0-1
is_confirmed	BooleanField	yes	User confirmed
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(profile, skill)

Indexes:

- profile
- skill
- source
- is_confirmed

Delete behavior:

- profile deleted -> profile skills deleted
- skill should not be deleted normally; use Skill.is_active=false

8. CV Schema

8.1 cvs.CVUpload

Stores CV file metadata. Physical file is stored in private media. This model uses soft delete, with a mandatory safe manager.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
public_id	UUIDField	yes	Unique public identifier
user	ForeignKey(User)	yes	Owner
file	FileField	yes	Private media path
original_filename	CharField	yes	Original upload name
file_hash	CharField	yes	SHA256
file_size	PositiveIntegerField	yes	Bytes
mime_type	CharField	yes	application/pdf
is_active	BooleanField	yes	One active CV
parse_status	CharField choices	yes	pending/processing/etc.
parse_error	TextField	no	Failure reason
text_extraction_status	CharField	no	success/failed/too_little_text
extracted_text_length	PositiveIntegerField	no	Number of chars
uploaded_at	DateTimeField	yes	Auto
parsed_at	DateTimeField	no	Set after parse
deleted_at	DateTimeField	no	Soft delete timestamp

Mandatory implementation:

```
class ActiveCVManager(models.Manager):
    def get_queryset(self):
        return super().get_queryset().filter(deleted_at__isnull=True)

class CVUpload(models.Model):
    objects = ActiveCVManager()      # safe default manager
    all_objects = models.Manager()   # admin/privacy/deletion/internal access

    public_id = models.UUIDField(default=uuid.uuid4, unique=True, editable=False, db_index=True)

    def soft_delete(self):
        self.is_active = False
        self.deleted_at = timezone.now()
        self.parse_status = "deleted"
```

```
self.save(update_fields=["is_active", "deleted_at", "parse_status"])
```

Constraints:

- unique(public_id)
- partial unique: unique(user) where is_active=true and deleted_at is null

Indexes:

- user
- public_id
- file_hash
- parse_status
- is_active
- uploaded_at
- deleted_at

8.2 cvs.CVParsedData

Stores parsed and structured CV extraction results. Includes deterministic extraction and LLM structured extraction.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
cv_upload	OneToOneField(CVUpload)	yes	Parsed data for CV
raw_text	TextField	no	Extracted PDF text
deterministic_json	JSONField	no	Regex extracted fields
llm_json	JSONField	no	Structured LLM output
merged_json	JSONField	no	Combined extraction
extraction_method	CharField	yes	regex_only/regex_llm/manual/failed
llm_schema_version	CharField	no	Example cv_extract_v1
extracted_name	CharField	no	Convenience field
extracted_email	EmailField	no	Convenience field
extracted_phone	CharField	no	Convenience field
extracted_location	CharField	no	Convenience field
extracted_linkedin_url	URLField	no	Convenience field
extracted_github_url	URLField	no	Convenience field
extracted_portfolio_url	URLField	no	Convenience field
estimated_years_experience	DecimalField	no	LLM/user estimated
confidence_json	JSONField	no	Field confidence
warnings_json	JSONField	no	Extraction warnings
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

LLM extraction schema contract:

```
{
  "candidate": {
    "full_name": null,
    "email": null,
    "phone": null,
    "location": null,
    "linkedin_url": null,
    "github_url": null,
    "portfolio_url": null,
    "website_url": null
  },
  "skills": {"technical": [], "tools": [], "soft": []},
  "experience": [{"title": null, "company": null, "start_date": null, "end_date": null, "description": null, "technologies": []}],
  "projects": [{"name": null, "description": null, "technologies": [], "links": []}],
  "education": [{"degree": null, "institution": null, "start_year": null, "end_year": null}],
  "certifications": [],
}
```

```

"languages": [{"language": null, "level": null}],
"estimated_years_experience": null,
"warnings": [],
"confidence": {"identity": 0, "skills": 0, "experience": 0, "education": 0, "languages": 0}
}

```

Indexes:

- cv_upload
- extraction_method
- llm_schema_version
- created_at

Delete behavior:

- CVUpload deleted -> CVParsedData deleted

9. Job Source and Ingestion Schema

9.1 jobs.JobSource

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
name	CharField	yes	Human-readable
slug	SlugField	yes	france_travail
base_url	URLField	no	API base
source_type	CharField	yes	api/manual/fixture
is_active	BooleanField	yes	Enable/disable
last_successful_sync_at	DateTimeField	no	Last success
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(slug)

Indexes:

- slug
- is_active
- last_successful_sync_at

MVP seed:

- slug = france_travail
- source_type = api

9.2 jobs.IngestionRun

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
source	ForeignKey(JobSource)	yes	Source
status	CharField choices	yes	running/success/partial_success/failed
trigger_type	CharField	yes	scheduled/manual_admin/startup_fixture/retry
started_at	DateTimeField	yes	Start time
finished_at	DateTimeField	no	End time
search_params_json	JSONField	no	Query params
fetched_count	PositiveIntegerField	yes	Default 0
created_count	PositiveIntegerField	yes	Default 0
updated_count	PositiveIntegerField	yes	Default 0
unchanged_count	PositiveIntegerField	yes	Default 0
marked_stale_count	PositiveIntegerField	yes	Default 0
marked_expired_count	PositiveIntegerField	yes	Default 0

Field	Type	Required	Notes
error_count	PositiveIntegerField	yes	Default 0
error_message	TextField	no	Summary
created_at	DateTimeField	yes	Auto

Indexes:

- source
- status
- started_at
- finished_at
- trigger_type

Delete behavior:

- source deleted -> protect

9.3 jobs.RawJobRecord

Stores original external API payload. Do not normalize external data without this record.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
source	ForeignKey(JobSource)	yes	Source
source_job_id	CharField	yes	External ID
raw_payload_json	JSONField	yes	Original payload
payload_hash	CharField	yes	SHA256 canonical JSON
first_seen_at	DateTimeField	yes	First import
last_seen_at	DateTimeField	yes	Last seen in source
last_fetched_at	DateTimeField	yes	Last fetched detail
source_status	CharField	no	Source-specific status
normalization_status	CharField	yes	pending/success/failed/skipped_unchanged
normalization_error	TextField	no	Error
ingestion_run	ForeignKey(IngestionRun)	no	Last run
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(source, source_job_id)

Indexes:

- source
- source_job_id
- payload_hash
- last_seen_at
- last_fetched_at
- normalization_status
- ingestion_run

Delete behavior:

- source deleted -> protect
- ingestion_run deleted -> set null

9.4 jobs.NormalizedJob

Stores clean searchable jobs used by the application.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
public_id	UUIDField	yes	Unique public route ID
source	ForeignKey(JobSource)	yes	Source

Field	Type	Required	Notes
raw_record	OneToOneField(RawJobRecord)	yes	Raw payload
source_job_id	CharField	yes	External ID duplicate for lookup
title	CharField	yes	Job title
company_name	CharField	no	Company
location	CharField	no	Full location label
country	CharField	yes	MVP France
city	CharField	no	City
department	CharField	no	French department
region	CharField	no	Region
contract_type	CharField	no	CDI/CDD/etc.
remote_type	CharField choices	yes	remote/hybrid/on_site/unknown
job_type	CharField choices	yes	internship/full_time/etc.
experience_level	CharField	yes	internship/junior/mid_level/senior/unknown
description	TextField	yes	Clean description
source_url	URLField	no	External apply/details link
published_at	DateTimeField	no	Source publication
expires_at	DateTimeField	no	Source expiry if available
first_seen_at	DateTimeField	yes	From raw
last_seen_at	DateTimeField	yes	From raw
last_fetched_at	DateTimeField	yes	From raw
status	CharField choices	yes	active/stale/etc.
normalization_version	CharField	yes	Example ft_v1
skill_extraction_status	CharField	yes	pending/success/failed/not_enough_text
required_skills_json	JSONField	no	Raw/canonical labels snapshot
optional_skills_json	JSONField	no	Raw/canonical labels snapshot
language_requirements_json	JSONField	no	French/English requirements
search_vector	SearchVectorField	no	PostgreSQL FTS
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(source, source_job_id)
- unique(raw_record)
- unique(public_id)

Indexes:

- public_id
- source
- source_job_id
- status
- job_type
- contract_type
- remote_type
- experience_level
- country, city, department, region
- published_at, expires_at
- last_seen_at, last_fetched_at
- search_vector GIN

Full-text search weights:

- title: A
- required skills: A
- optional skills: B
- company: B

- location: C
- description: D

Delete behavior:

- raw_record deleted -> normalized job deleted
- source deleted -> protect
- normal operational deletion should be status update, not delete

9.5 jobs.NormalizedJobSkill

Stores canonical skills attached to a job. Matching uses this relational table.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
job	ForeignKey(NormalizedJob)	yes	Job
skill	ForeignKey(Skill)	yes	Canonical skill
requirement_type	CharField	yes	required/optional/detected/unknown
source	CharField	yes	rule/llm/admin/source_api
confidence	DecimalField	no	0-1
created_at	DateTimeField	yes	Auto

Constraints:

- unique(job, skill, requirement_type)

Indexes:

- job
- skill
- requirement_type
- source
- confidence

Delete behavior:

- job deleted -> job skills deleted
- skill deleted -> protect

10. Skill Taxonomy Schema

10.1 skills.Skill

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
canonical_name	CharField	yes	Python, Django, Docker
slug	SlugField	yes	python
category	CharField choices	yes	backend/devops/etc.
is_active	BooleanField	yes	Use false instead of delete
source	CharField	yes	manual/esco/imported/admin
esco_uri	URLField	no	Optional ESCO reference
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Categories:

programming_language, frontend, backend, database, devops, cloud, testing, data_ai, mobile, tools, methodology, soft_skill, other

Constraints:

- unique(slug)
- unique(canonical_name)

Indexes:

- slug
- canonical_name

- category
- is_active
- source

Delete behavior:

- Do not delete skills normally. Use is_active=false.

10.2 skills.SkillAlias

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
skill	ForeignKey(Skill)	yes	Canonical skill
alias	CharField	yes	Original alias
normalized_alias	CharField	yes	Lowercase normalized form
language	CharField	no	fr/en/unknown
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Examples:

- ReactJS -> React
- Postgres -> PostgreSQL
- Python3 -> Python

Constraints:

- unique(normalized_alias)

Indexes:

- skill
- normalized_alias
- language

Delete behavior:

- skill deleted -> protect

10.3 skills.UnmatchedSkillCandidate

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
raw_skill_text	CharField	yes	Original text
normalized_text	CharField	yes	Normalized text
source_type	CharField	yes	cv/job/quick_match/manual
source_model	CharField	no	CVUpload/NormalizedJob
source_object_id	BigIntegerField	no	Source ID
occurrence_count	PositiveIntegerField	yes	Default 1
status	CharField choices	yes	pending/mapped/ignored
mapped_skill	ForeignKey(Skill)	no	If mapped
reviewed_by	ForeignKey(User)	no	Admin
reviewed_at	DateTimeField	no	Review time
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(normalized_text, source_type)

Indexes:

- normalized_text
- source_type
- status
- mapped_skill
- reviewed_by

- occurrence_count
- created_at

Delete behavior:

- mapped_skill deleted -> set null
- reviewed_by deleted -> set null

11. Matching Schema

11.1 matching.MatchResult

Stores a full match calculation between a user/profile/CV and a job. Includes snapshots to preserve past match context.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
public_id	UUIDField	yes	Public-safe ID
user	ForeignKey(User)	yes	Owner
profile	ForeignKey(CandidateProfile)	yes	Profile snapshot source
cv_upload	ForeignKey(CVUpload)	no	Active CV used
job	ForeignKey(NormalizedJob)	yes	Matched job
profile_snapshot_json	JSONField	yes	Profile at calculation time
job_snapshot_json	JSONField	yes	Job at calculation time
fit_score	PositiveSmallIntegerField	yes	0-100
technical_skills_score	PositiveSmallIntegerField	yes	0-100
experience_score	PositiveSmallIntegerField	yes	0-100
role_title_score	PositiveSmallIntegerField	yes	0-100
language_score	PositiveSmallIntegerField	yes	0-100
location_score	PositiveSmallIntegerField	yes	0-100
strong_skills_json	JSONField	no	Matched skills
missing_required_skills_json	JSONField	no	Missing required
missing_optional_skills_json	JSONField	no	Missing optional
risk_flags_json	JSONField	no	Score-relevant risks
profile_signals_json	JSONField	no	Non-score profile signals
recommended_actions_json	JSONField	no	Deterministic actions
llm_explanation_status	CharField choices	yes	not_requested/etc.
llm_explanation_text	TextField	no	AI explanation
scoring_version	CharField	yes	Example score_v1
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Fit score formula:

```
fit_score =
    technical_skills_score * 0.45
+ experience_score * 0.20
+ role_title_score * 0.15
+ language_score * 0.10
+ location_score * 0.10
```

No unique(user, job): user may recompute after CV/profile changes.

Indexes:

- public_id

- user, profile, cv_upload, job
- fit_score
- created_at
- llm_explanation_status
- scoring_version
- (user, job)
- (user, created_at)
- (job, fit_score)

11.2 matching.QuickMatchSession

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
public_id	UUIDField	yes	Public-safe ID
session_key_hash	CharField	yes	Hashed session key
ip_hash	CharField	no	For rate limiting
job	ForeignKey(NormalizedJob)	yes	Tested job
entered_skills_json	JSONField	yes	User-entered skills
normalized_skills_json	JSONField	no	Canonical skills
experience_level	CharField	yes	User input
french_level	CharField	yes	User input
estimated_fit_score	PositiveSmallIntegerField	yes	0-100
matched_skills_json	JSONField	no	Matched
missing_skills_json	JSONField	no	Missing
created_at	DateTimeField	yes	Auto
expires_at	DateTimeField	yes	24h default

Constraints:

- unique(public_id)

Indexes:

- public_id
- session_key_hash
- ip_hash
- job
- created_at
- expires_at
- estimated_fit_score

12. Recommendation Schema

12.1 recommendations.JobRecommendation

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	yes	Owner
profile	ForeignKey(CandidateProfile)	yes	Profile used
cv_upload	ForeignKey(CVUpload)	no	CV used
job	ForeignKey(NormalizedJob)	yes	Recommended job
fit_score	PositiveSmallIntegerField	yes	Candidate-job fit
ranking_score	DecimalField	yes	Final rank
rank	PositiveIntegerField	yes	1, 2, 3...
strong_skills_json	JSONField	no	Matched
missing_skills_json	JSONField	no	Missing
risk_flags_json	JSONField	no	Risks

Field	Type	Required	Notes
profile_signals_json	JSONField	no	Non-score signals
reason_summary	TextField	no	Short deterministic summary
recommendation_version	CharField	yes	reco_v1
computed_at	DateTimeField	yes	Calculation time
status	CharField choices	yes	active/stale/dismissed/expired_job
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Ranking score:

$\text{ranking_score} = \text{fit_score} + \text{freshness_boost} + \text{target_type_boost} + \text{user_preference_boost} - \text{stale_job_penalty}$

Constraints:

- unique(user, job, recommendation_version) for MVP

Indexes:

- user, profile, cv_upload, job
- fit_score, ranking_score, rank
- status, computed_at, recommendation_version
- (user, status, ranking_score)
- (user, rank)
- (job, status)

12.2 recommendations.RecommendationRun

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	no	Null for bulk run
trigger_type	CharField	yes	cv_uploaded/profile_updated/etc.
status	CharField choices	yes	running/success/partial_success/failed
started_at	DateTimeField	yes	Start
finished_at	DateTimeField	no	End
candidate_jobs_count	PositiveIntegerField	yes	Default 0
scored_jobs_count	PositiveIntegerField	yes	Default 0
stored_recommendations_count	PositiveIntegerField	yes	Default 0
error_message	TextField	no	Error
created_at	DateTimeField	yes	Auto

Trigger types:

cv_uploaded, cv_replaced, profile_updated, new_jobs_imported, dashboard_stale_refresh, nightly_refresh, manual_admin

Indexes:

- user
- trigger_type
- status
- started_at
- finished_at

Delete behavior:

- user deleted -> set null

12.3 recommendations.SavedJob

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	yes	Owner

Field	Type	Required	Notes
job	ForeignKey(NormalizedJob)	yes	Saved job
notes	TextField	no	Future
saved_at	DateTimeField	yes	Auto

Constraints:

- unique(user, job)

Indexes:

- user
- job
- saved_at

13. LLM Schema

13.1 IIm.PromptVersion

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
name	CharField	yes	cv_extract
version	CharField	yes	v1
purpose	CharField	yes	cv_extraction/etc.
template_text	TextField	yes	Prompt
output_schema_json	JSONField	no	Expected schema
is_active	BooleanField	yes	Active version
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Purposes:

cv_extraction, job_skill_extraction, match_explanation, cv_advice, missing_skill_roadmap

Constraints:

- unique(name, version)

Indexes:

- name
- purpose
- is_active
- created_at

13.2 IIm.LLMCacheEntry

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
cache_key	CharField	yes	Unique
purpose	CharField	yes	cv_extraction/etc.
input_hash	CharField	yes	Hash of input
prompt_version	ForeignKey(PromptVersion)	yes	Prompt used
response_json	JSONField	no	JSON result
response_text	TextField	no	Raw text
status	CharField	yes	success/failed
created_at	DateTimeField	yes	Auto
expires_at	DateTimeField	no	Optional

Constraints:

- unique(cache_key)

Indexes:

- cache_key
- purpose
- input_hash
- prompt_version
- status
- created_at
- expires_at

Delete behavior:

- prompt_version deleted -> protect

13.3 llm.LLMUsageLog

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	no	Null for system jobs
purpose	CharField	yes	cv_extraction/etc.
provider	CharField	yes	openrouter
model	CharField	yes	Model name
prompt_version	ForeignKey(PromptVersion)	no	Prompt
cache_hit	BooleanField	yes	True/false
input_token_estimate	PositiveIntegerField	no	Optional
output_token_estimate	PositiveIntegerField	no	Optional
cost_estimate	DecimalField	no	Optional
status	CharField	yes	success/failed/cached/skipped_limit
error_message	TextField	no	Error
created_at	DateTimeField	yes	Auto

Indexes:

- user
- purpose
- provider
- model
- cache_hit
- status
- created_at
- prompt_version

Delete behavior:

- user deleted -> set null
- prompt_version protected

14. Notification Schema

14.1 notifications.EmailPreference

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	OneToOneField(User)	yes	One preferences row
weekly_digest_enabled	BooleanField	yes	Default false
product_updates_enabled	BooleanField	yes	Default false
instant_alerts_enabled	BooleanField	yes	Default false, hidden MVP
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Constraints:

- unique(user)

Indexes:

- user
- weekly_digest_enabled
- product_updates_enabled
- instant_alerts_enabled

14.2 notifications.EmailBatch

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
batch_type	CharField	yes	weekly_digest/product_update/system_notice
period_start	DateTimeField	yes	Week start
period_end	DateTimeField	yes	Week end
status	CharField	yes	running/success/partial_success/failed
created_at	DateTimeField	yes	Auto
completed_at	DateTimeField	no	End time
error_message	TextField	no	Error

Constraints:

- unique(batch_type, period_start, period_end)

Indexes:

- batch_type
- status
- period_start
- period_end
- created_at

14.3 notifications.EmailEvent

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	no	Recipient user
batch	ForeignKey(EmailBatch)	no	For digest/batch
email_type	CharField	yes	verification/digest/etc.
recipient_email	EmailField	yes	Recipient
subject	CharField	yes	Subject
provider	CharField	yes	resend/brevo/etc.
provider_message_id	CharField	no	Provider ID
idempotency_key	CharField	yes	Unique send key
status	CharField	yes	queued/sent/failed/skipped
error_message	TextField	no	Failure
sent_at	DateTimeField	no	Send time
created_at	DateTimeField	yes	Auto

Constraints:

- unique(idempotency_key)

Example idempotency keys:

- weekly_digest:user_id:2026-W24
- email_verification:user_id:token_hash
- password_reset:user_id:token_hash

Indexes:

- user, batch, email_type, recipient_email, provider, provider_message_id
- idempotency_key
- status
- sent_at
- created_at

14.4 notifications.EmailUnsubscribeToken

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	yes	User
token_hash	CharField	yes	Store hash, not raw token
purpose	CharField	yes	digest/product_updates
created_at	DateTimeField	yes	Auto
expires_at	DateTimeField	no	Optional
used_at	DateTimeField	no	Optional

Constraints:

- unique(token_hash)

Indexes:

- user
- token_hash
- purpose
- expires_at
- used_at

15. Privacy Schema

15.1 privacy.ConsentRecord

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	yes	User
consent_type	CharField	yes	cv_processing/etc.
consent_text	TextField	yes	Text accepted
consent_version	CharField	yes	Version
accepted_at	DateTimeField	yes	Acceptance time
ip_address_hash	CharField	no	Hashed IP
user_agent	TextField	no	Browser info

Consent types:

- cv_processing
- email_digest
- terms
- privacy_policy
- cookie_session_notice

Indexes:

- user
- consent_type
- consent_version
- accepted_at

15.2 privacy.DeletionRequest

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	no	Set null after deletion if needed
status	CharField	yes	pending/processing/completed/failed
requested_at	DateTimeField	yes	Request time
processed_at	DateTimeField	no	Completion time
error_message	TextField	no	Error

Field	Type	Required	Notes
metadata_json	JSONField	no	Counts deleted/anonymized

Indexes:

- user
- status
- requested_at
- processed_at

Delete behavior:

- user deleted -> set null

16. Analytics Schema

16.1 analytics.UserEvent

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
user	ForeignKey(User)	no	Null for anonymous
session_key_hash	CharField	no	Anonymous tracking
event_type	CharField	yes	Event name
metadata_json	JSONField	no	Extra context
created_at	DateTimeField	yes	Auto

Event types:

job_search, job_detail_view, quick_match_started, quick_match_completed, signup_started, signup_completed, cv_uploaded, cv_parse_failed, profile_completed, match_generated, recommendation_clicked, saved_job_created, source_apply_clicked, email_digest_clicked, account_deleted

Indexes:

- user
- session_key_hash
- event_type
- created_at

Delete behavior:

- user deleted -> set null and remove personal metadata during deletion

17. Optional Core Schema

17.1 core.SystemSetting

Recommended early because it avoids hardcoding thresholds.

Field	Type	Required	Notes
id	BigAutoField	yes	Primary key
key	CharField	yes	Unique
value_json	JSONField	yes	Setting value
description	TextField	no	Admin help
is_active	BooleanField	yes	Active
created_at	DateTimeField	yes	Auto
updated_at	DateTimeField	yes	Auto

Example keys:

- job_stale_after_hours
- job_removed_after_hours
- quick_match_daily_limit
- llm_daily_limit_free_user
- recommendation_refresh_hours

Constraints:

- unique(key)

Indexes:

- key
- is_active

18. Full Relationship Map

User

```
|-- CandidateProfile
|   |-- ProfileSkill -> Skill
|-- CVUpload
|   |-- CVParsedData
|-- MatchResult -> NormalizedJob
|-- JobRecommendation -> NormalizedJob
|-- SavedJob -> NormalizedJob
|-- EmailPreference
|-- EmailEvent
|-- LLMUsageLog
|-- ConsentRecord
|-- DeletionRequest
|-- UserEvent
```

JobSource

```
|-- IngestionRun
|-- RawJobRecord
|   |-- NormalizedJob
|       |-- NormalizedJobSkill -> Skill
|-- NormalizedJob
```

Skill

```
|-- SkillAlias
|-- ProfileSkill
|-- NormalizedJobSkill
|-- UnmatchedSkillCandidate(mapped_skill)
```

PromptVersion

```
|-- LLMCacheEntry
|-- LLMUsageLog
```

EmailBatch

```
|-- EmailEvent
```

19. Index Strategy Summary

19.1 Critical Indexes

User.email unique
CandidateProfile.user unique
CVUpload.user + is_active partial unique where deleted_at is null
RawJobRecord.source + source_job_id unique
NormalizedJob.source + source_job_id unique
NormalizedJob.public_id unique
NormalizedJob.search_vector GIN
Skill.slug unique
SkillAlias.normalized_alias unique
ProfileSkill.profile + skill unique
NormalizedJobSkill.job + skill + requirement_type unique
SavedJob.user + job unique
EmailEvent.idempotency_key unique
LLMCacheEntry.cache_key unique

19.2 Worker/Admin Indexes

IngestionRun.status + started_at
RawJobRecord.normalization_status
CVUpload.parse_status

```

UnmatchedSkillCandidate.status
LLMUsageLog.status + created_at
EmailEvent.status + created_at
DeletionRequest.status

```

20. JSON Field Strategy

Use JSONField for flexible nested data, snapshots, raw payloads, LLM output, warnings, and metadata. Use relational tables for searched, filtered, joined, counted, and scored data.

Use JSONField For	Use Relational Tables For
RawJobRecord.raw_payload_json	User
IngestionRun.search_params_json	CandidateProfile
CVParsedData.deterministic_json / llm_json / merged_json	CVUpload
CVParsedData.confidence_json / warnings_json	NormalizedJob
MatchResult.profile_snapshot_json / job_snapshot_json	Skill and SkillAlias
MatchResult strong/missing/risk/profile/action JSON	ProfileSkill and NormalizedJobSkill
JobRecommendation strong/missing/risk/profile JSON	MatchResult and JobRecommendation
UserEvent.metadata_json	EmailEvent

21. Deletion and Privacy Behavior

21.1 Delete CV

1. Remove physical file.
2. Set CVUpload.deleted_at.
3. Set parse_status=deleted.
4. Delete or clear CVParsedData.
5. Remove ProfileSkill rows sourced only from that CV if not confirmed.
6. Mark related recommendations stale.

21.2 Replace CV

1. Existing active CV becomes inactive.
2. New CVUpload created as active.
3. Parse task queued.
4. Profile review required.
5. Recommendations refreshed after confirmation.

21.3 Delete Account

1. Create DeletionRequest.
2. Disable user login.
3. Delete CV files.
4. Delete CVParsedData.
5. Delete CVUpload rows or anonymize.
6. Delete CandidateProfile.
7. Delete ProfileSkill.
8. Delete SavedJob.
9. Delete MatchResult.
10. Delete JobRecommendation.
11. Delete EmailPreference.
12. Delete social accounts.
13. Remove/anonymize UserEvent personal metadata.
14. Delete or anonymize User.
15. Mark DeletionRequest completed.

MVP recommendation:
 Hard-delete user-owned candidate data.
 Keep anonymized operational logs only where useful.

22. Migration Order

Batch	Models
1. Foundation	accounts.User; core.SystemSetting optional
2. Skills	skills.Skill; skills.SkillAlias; skills.UnmatchedSkillCandidate
3. Profiles	profiles.CandidateProfile; profiles.ProfileSkill
4. Jobs	jobs.JobSource; jobs.IngestionRun; jobs.RawJobRecord; jobs.NormalizedJob; jobs.NormalizedJobSkill
5. CVs	cvs.CVUpload; cvs.CVParsedData
6. Matching	matching.MatchResult; matching.QuickMatchSession
7. Recommendations	recommendations.JobRecommendation; RecommendationRun; SavedJob
8. LLM	llm.PromptVersion; LLMCacheEntry; LLMUsageLog
9. Notifications	notifications.EmailPreference; EmailBatch; EmailEvent; EmailUnsubscribeToken
10. Privacy and Analytics	privacy.ConsentRecord; DeletionRequest; analytics.UserEvent

23. Seed Data Requirements

23.1 Required Seed Data

- JobSource: france_travail
- Skill categories
- Initial 200-300 skills
- Initial aliases
- System settings
- Prompt versions

23.2 Initial System Settings

```
job_stale_after_hours = 24
job_removed_after_hours = 72
job_revalidate_after_hours = 6
recommendation_stale_after_hours = 24
active_user_days = 14
quick_match_expiry_hours = 24
max_cv_upload_size_mb = 5
max_quick_matches_per_ip_day = 10
max_llm_calls_per_user_day = 10
max_match_explanations_per_user_day = 5
```

23.3 Initial Prompt Versions

```
cv_extraction:v1
job_skill_extraction:v1
match_explanation:v1
cv_advice:v1
```

24. MVP Tables List

```
accounts_user
profiles_candidateprofile
profiles_profileskill

cvs_cvupload
cvs_cvparseddata

jobs_jobsource
jobs_ingestionrun
jobs_rawjobrecord
jobs_normalizedjob
```

jobs_normalizedjobskill

skills_skill
skills_skillalias
skills_unmatchedskillcandidate

matching_matchresult
matching_quickmatchsession

recommendations_jobrecommendation
recommendations_recommendationrun
recommendations_savedjob

notifications_emailpreference
notifications_emailbatch
notifications_emailevent
notifications_emailunsubscribe token

llm_promptversion
llm_llmcacheentry
llm_llmusagelog

privacy_consentrecord
privacy_deletionrequest

analytics_userevent

Optional but recommended:
core_systemsetting

25. Implementation Safety Notes

25.1 Soft-deleted CVs

CVUpload must use a default manager that excludes deleted_at records.
A secondary all_objects manager is required for admin/privacy/deletion operations.
Normal app code uses CVUpload.objects.
Deletion/privacy/admin code uses CVUpload.all_objects.
Do not soft-delete everything. Use soft delete only where it has value.

25.2 Target roles

CandidateProfile.target_roles remains JSONField for MVP.
This is acceptable because recommendation workers usually load one user profile and compare in memory.
If role matching or analytics become slow, migrate later to:

- TargetRole
- CandidateProfileTargetRole

25.3 Public IDs

All public-facing models must use UUID public_id with:

- default=uuid.uuid4
- unique=True
- editable=False
- db_index=True

Public routes must use public_id, never internal pk.

26. Schema Risks

Too Many JSON Fields

Mitigation: Use relational tables for skills, job skills, profile skills, recommendations, match scores, and email events. Use JSON only for snapshots, raw payloads, LLM output, warnings, and metadata.

Skill Taxonomy Pollution

Mitigation: Admin review for unmatched skills. Unique normalized aliases. Never auto-create Skill records directly from LLM output.

Recommendation Staleness

Mitigation: Mark recommendations stale after profile/CV update. Store recommendation_version and computed_at. Refresh via Celery.

Privacy Deletion Complexity

Mitigation: Central DeletionRequest worker. Clear deletion order. Private CV storage. No public CV URLs.

Search Vector Maintenance

Mitigation: Update search_vector during normalization/save. Rebuild after skill extraction updates. Add management command to rebuild search index.

27. Database Schema Acceptance Criteria

- User/auth data model is clear.
- Candidate profile model is clear.
- CV upload and parsing model is clear.
- Raw and normalized job models are clear.
- France Travail source support is clear.
- Skill taxonomy and aliases are defined.
- Unknown skill review is supported.
- PostgreSQL full-text search is supported.
- Match result storage is defined.
- Recommendation storage is defined.
- Email digest idempotency is defined.
- LLM cache and usage logging are defined.
- Consent and deletion flows are supported.
- Product event logging is supported.
- Indexes and constraints are defined.
- Migration order is clear.
- Soft-delete manager requirements are documented.
- Public UUID routing is documented.

28. Final Schema Decision

Build the database as a normalized Django/PostgreSQL schema with controlled JSON usage.

Users and profiles are relational.
Jobs are raw + normalized.
Skills are canonical relational entities.
Matching results are stored with score breakdowns.
Recommendations are precomputed and stored.
LLM output is cached and logged.
Privacy deletion is first-class.
Search uses PostgreSQL full-text search.
Public URLs use UUID public_id, not internal integer IDs.
Soft-deleted CV records are hidden by default manager.

Next document after validation: Service Contracts v1.